∀ for all
∃ exist, some

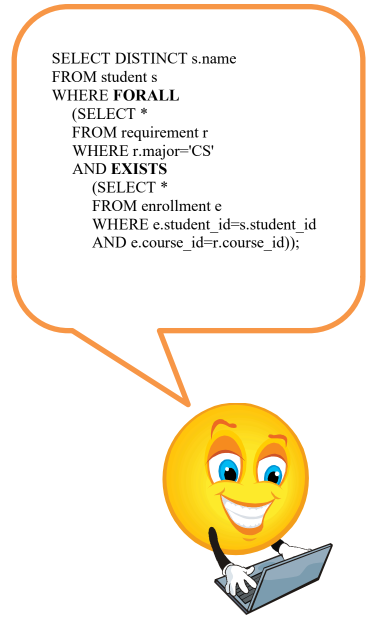
think ∀ everybody
∃ somebody

x Loves y

COMP 2121 DISCRETE MATHEMATICS | LAB 4

1. Determine the truth value of the following statements (T or F) and justify your answer. (hint: you can use counterexample to disprove the statement)

   The universe for x is numbers: 8, 6, 4, 2 and 0.
   The universe for y is numbers: −7, −5, −3, and −1.



a) $\exists x \exists y$ *such that* $-30 < x \cdot y < -5$

x	y	$-30 < xy < -5$
4	-5	$-30 < -20 < -5$ ✓

TRUE

copy the whole statement, sub the numbers

b) $\forall x \forall y$ *such that* $x \cdot y < x + y$

x	y	$xy < x+y$
0	-1	$0 < -1$ ×

FALSE

c) $\forall x \exists y$ *such that* $x + y = 1$

x	y	$x+y=1$
0	-7	$-7=1$ ×
0	-5	$-5=1$ ×
0	-3	$-3=1$ ×
0	-1	$-1=1$ ✓

FALSE

d) $\forall y \exists x$ *such that* $x + y = 1$

y	x	$x+y=1$
-7	8	$1=1$ ✓
-5	6	$1=1$ ✓
-3	4	$1=1$ ✓
-1	2	$1=1$ ✓

TRUE

e) $\exists x \forall y$ *such that* $x \cdot y > -10$

x	y	$xy > -10$
0	-7	$0 > -10$ ✓
0	-5	$0 > -10$ ✓
0	-3	$0 > -10$ ✓
0	-1	$0 > -10$ ✓

TRUE

2. Determine the truth value of the following statements (T or F) and justify your answer.

Sets A, B and C are defined as follows:

$\mathrm{A} = \{1, 2, 3\} \qquad \mathrm{B} = \{-1, -2, -3\} \qquad \mathrm{C} = \{-1, -2, -4, -9\}.$

a) $\forall\, x \in \mathrm{A}, \forall\, y \in B,$ then $x \cdot y \in C$

x	y	xy∈C
-1	-3	-3∈C ×

FALSE

b) $\forall\, x \in \mathrm{A}, \exists\, y \in \mathrm{B},$ then $x \cdot y \in C$

x	y	xy∈C
1	-1	-1 ∈C ✓
2	-2	-4 ∈C ✓
3	-3	-9 ∈C ✓

TRUE

c) $\exists\, y \in \mathrm{B}, \forall\, x \in \mathrm{A},$ then $x \cdot y \in C$

y	x	xy∈C
-1	3	-3 ∈C ×
-2	3	-6 ∈C ×
-3	1	-3 ∈C ×

FALSE

d) $\exists z \in \mathrm{C}, \forall y \in \mathrm{B}$ such that $z \cdot y \notin \mathrm{A}$

z	y	zy∉A
-9	-1	9 ∉A ✓
-9	-2	18 ∉A ✓
-9	-3	27 ∉A ✓

TRUE

e) $\exists u \in \mathrm{A}, \exists v \in \mathrm{A}$ such that $u \cdot v \in \mathrm{C}$

u	v	uv∈C
1	1	1∈C ×
1	2	2∈C ×
1	3	3∈C ×
2	1	2∈C ×
2	2	4∈C ×
2	3	6∈C ×
3	1	3∈C ×
3	2	6∈C ×
3	3	9∈C ×

FALSE

3. Determine the truth value of the following statements (T or F) and justify your answer.

The universe for x is {-4, -2, -1}.
The universe for y is {-4, -2, 4, 6}

(a) $\forall x, \forall y,\ \ 0 < |x - y| < 6$

x	y	$0 < \lvert x-y \rvert < 6$
-4	-4	$0 < 0 < 6$ ×

FALSE

(b) $\exists x, \exists y,\ \ 0 < |x - y| < 6$

x	y	$0 < \lvert x-y \rvert < 6$
-4	-2	$0 < 2 < 6$ ✓

TRUE

(c) $\forall x, \exists y,\ \ 0 < |x - y| < 6$

x	y	$0 < \lvert x-y \rvert < 6$
-4	-2	$0 < 2 < 6$ ✓
-2	-1	$0 < 1 < 6$ ✓
-1	-2	$0 < 1 < 6$ ✓

TRUE

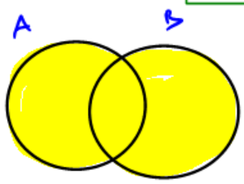
(d) $\exists x, \forall y,\ \ 0 < |x - y| < 6$

x	y	$0 < \lvert x-y \rvert < 6$
-4	-4	$0 < 0 < 6$ ×
-2	-2	$0 < 0 < 6$ ×
-1	6	$0 < 7 < 6$ ×

FALSE

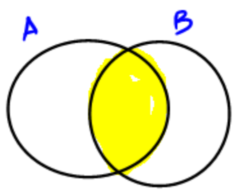
4. Determine the following:

a) $X \cup Y \cup Z = \{1, 2, 3, 4, 6, 7\}$



$A \cup B$

b) $Y \cap Z = \{\ \}$



$A \cap B$

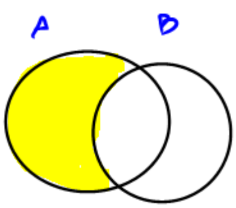
X = {1, 2, 3, 4}

Y = {1, 3, 7}

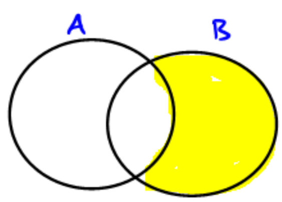
Z = {4, 6} and

universe U = {1, 2, 3, 4, 5, 6, 7, 8}

c) $Y - (X \cup Z) = \{1,3,7\} - \{1,2,3,4,6\} = \{7\}$

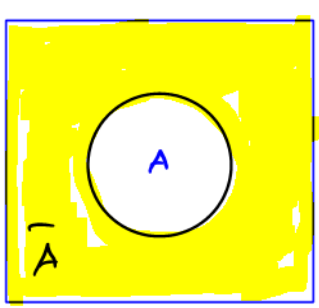


$A - B$

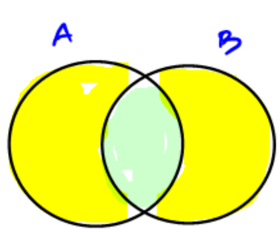


$B - A$

d) $\overline{(X - Z)} \Delta Y = \overline{\{1,2,3\}} \Delta \{1,3,7\} = \{4,5,6,7,8\} \Delta \{1,3,7\} = \{1,3,4,5,6,8\}$



$\bar{A}$



$A \Delta B$

Determine the truth value of the following statements (T or F) and justify your answer.

X = {1, 2, 3, 4}
Y = {1, 3, 7}
Z = {4, 6} and
universe U = {1, 2, 3, 4, 5, 6, 7, 8}

e) $\exists p \in U, \forall q \in \overset{\{2,4,7\}}{\cancel{X \Delta Y}}$, then $1 \le |p-q| \le 3$

p	q	$1 \le \lvert p-q \rvert \le 3$
5	2	$1 \le 3 \le 3$ ✓
5	4	$1 \le 1 \le 3$ ✓
5	7	$1 \le 2 \le 3$ ✓

$\boxed{\text{TRUE}}$

$X \Delta Y = \{2,4,7\}$

$X \cup Y = \{1,2,3,4,7\}$

$\overline{X \cup Y} = \{5,6,8\}$

$X - Z = \{1,2,3\}$

$X \cap Y = \{1,3\}$

$\overline{X \cap Y} = \{2,4,5,6,7,8\}$

Δ

$\{4,6\}$

$= \{2,5,7,8\}$

f) $\forall t \in \overset{\{5,6,8\}}{\cancel{\overline{X \cup Y}}}, \exists w \in \overset{\{1,2,3\}}{\cancel{X - Z}}$, then $t - w \in \overset{\{2,5,7,8\}}{\cancel{\overline{X \cap Y} \Delta Z}}$

t	w	$t - w \in \{2,5,7,8\}$
5	3	$2 \in \{2,5,7,8\}$ ✓
6	1	$5 \in \{2,5,7,8\}$ ✓
8	3	$5 \in \{2,5,7,8\}$ ✓

$\boxed{\text{TRUE}}$

g) $\exists a \in \overset{\{1,2,3\}}{\cancel{X - Z}}, \forall b \in \overset{\{2,5,7,8\}}{\cancel{\overline{X \cap Y} \Delta Z}}$, then $-a + b \notin \overset{\{2,5,7,8\}}{\cancel{\overline{X \cap Y} \Delta Z}}$

a	b	$-a + b \notin \{2,5,7,8\}$
1	8	$7 \notin \{2,5,7,8\}$ ×
2	7	$5 \notin \{2,5,7,8\}$ ×
3	5	$2 \notin \{2,5,7,8\}$ ×

$\boxed{\text{FALSE}}$